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# Install CoolProp if it is missing (e.g., on Google Colab)
try:
    import CoolProp
except ImportError:
    %pip install -q CoolProp

import CoolProp.CoolProp as CP

def isentropic_expansion(fluid, P1_bar, T1_C, P2_bar):
    """
    Calculates the isentropic exit state for a
    ↪ turbine/compressor.
    """
    print(f"--- ISENTROPIC PROCESS SOLVER ---")
    print(f"Fluid: {fluid}")

    # Convert inputs to SI units (Pa and K) for CoolProp
    P1_Pa = P1_bar * 1e5
    T1_K = T1_C + 273.15
    P2_Pa = P2_bar * 1e5

    # 1. Calculate State 1 (Inlet) Properties
    h1 = CP.PropsSI('H', 'P', P1_Pa, 'T', T1_K, fluid) / 1000 #
    ↪ Convert J/kg to kJ/kg
    s1 = CP.PropsSI('S', 'P', P1_Pa, 'T', T1_K, fluid) / 1000 #
    ↪ Convert J/kg-K to kJ/kg-K

    print(f"\nState 1 (Inlet):")
    print(f"  P1 = {P1_bar} bar")
    print(f"  T1 = {T1_C} °C")
    print(f"  h1 = {h1:.2f} kJ/kg")
    print(f"  s1 = {s1:.4f} kJ/kg-K")

    # 2. Define State 2 (Exit) via Isentropic Constraint (s2 =
    ↪ s1)
    s2 = s1
    s2_J = s2 * 1000 # Convert back to standard SI for CoolProp

    # 3. Calculate State 2 Properties
    T2_K = CP.PropsSI('T', 'P', P2_Pa, 'S', s2_J, fluid)
    T2_C = T2_K - 273.15
    h2 = CP.PropsSI('H', 'P', P2_Pa, 'S', s2_J, fluid) / 1000

    # Check phase/quality (CoolProp returns -1 for subcooled,
    ↪ values > 1 for superheated)
    x2 = CP.PropsSI('Q', 'P', P2_Pa, 'S', s2_J, fluid)

    print(f"\nState 2 (Ideal Isentropic Exit):")

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print(f"  P2 = {P2_bar} bar")
print(f"  s2 = {s2:.4f} kJ/kg-K  <-- (s2 = s1)")
print(f"  T2 = {T2_C:.2f} °C")
print(f"  h2 = {h2:.2f} kJ/kg")

if 0 <= x2 <= 1:
    print(f"  Phase: Two-Phase Liquid-Vapor Mixture")
    print(f"  Quality (x) = {x2:.4f}")
elif x2 < 0:
    print(f"  Phase: Subcooled Liquid")
else:
    print(f"  Phase: Superheated Vapor")

# Calculate ideal work (W_rev = \Delta H)
w_rev = h2 - h1
print(f"\nIdeal Turbine Work (W_rev): {w_rev:.2f} kJ/kg")

return h1, h2, w_rev

# --- Run the HP Turbine Example from Lecture 19 ---
# Inlet: 60 bar, 500 C. Outlet: 10 bar.
h_in, h_out, work = isentropic_expansion('Water', P1_bar=60,
    ↪ T1_C=500, P2_bar=10)

```

--- ISENTROPIC PROCESS SOLVER ---

Fluid: Water

State 1 (Inlet):

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P1 = 60 bar
T1 = 500 °C
h1 = 3423.11 kJ/kg
s1 = 6.8826 kJ/kg-K

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State 2 (Ideal Isentropic Exit):

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P2 = 10 bar
s2 = 6.8826 kJ/kg-K  <-- (s2 = s1)
T2 = 239.78 °C
h2 = 2920.40 kJ/kg
Phase: Subcooled Liquid

```

Ideal Turbine Work (W_rev): -502.71 kJ/kg